

Testing the Efficient Market Hypothesis: Do Financial Indicators Predict Stock Returns?

**ECO348-FAC413-DOM637 – Introduction to
Mathematical Finance and Financial Engineering**

Siddarth V – 2410111040

R.V Akilan – 2410110949

Sayantan Soundappan R - 2410111002

1. Introduction:

According to the Efficient Market Hypothesis (EMH), stock prices incorporate all publicly available information, implying that future returns cannot be predicted using information that is already known. Testing this hypothesis helps assess market efficiency.

This study examines whether firm-specific financial characteristics observed in 2014 are related to stock returns over the period 2014 - 2018. The variables considered include measures of profitability, earnings, financial leverage (dkr) , and CEO total compensation (salary). The dataset includes 142 firms and reflects pre-COVID market conditions.

If these financial indicators are found to significantly explain stock returns, it would suggest some degree of predictability, indicating that markets may not be fully efficient. Conversely, the absence of statistical significance would support the EMH.

2. Methodology:

2.1 Data:

This study uses a cross-sectional dataset of 142 firms. The dependent variable is the percentage change in stock price from end-2014 to end-2018 (return). The explanatory variables, all measured at end-2014 and publicly available at the beginning of the period, are:

- Return on equity (roe)
- Return on capital (roc)
- Debt-to-capital ratio (dkr)
- Earnings per share (eps)
- Net income (netinc, in millions ₹)
- CEO total compensation (sal, in thousands ₹)

These variables capture key aspects of firm performance, including profitability, financial leverage, earnings, and CEO compensation.

2.2 Model Specification:

The baseline model is estimated using Ordinary Least Squares (OLS):

$$\text{return} = \beta_0 + \beta_1 \text{roe} + \beta_2 \text{roc} + \beta_3 \text{dkr} + \beta_4 \text{eps} + \beta_5 \text{netinc} + \beta_6 \text{sal} + u$$

where u is the error term. The significance level for all tests is set at 5%.

2.3 Hypotheses:

Joint hypothesis:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0$$

$$H_1: \text{At least one } \beta_i \neq 0$$

Individual hypotheses (for each variable i)

$$H_0: \beta_i = 0$$

$$H_1: \beta_i \neq 0$$

2.4 Analysis:

The analysis proceeds in two steps. First, the baseline OLS regression model is estimated. The overall significance of the model is evaluated using the F-test, while individual coefficients are assessed using t-tests based on p-values.

Second, the distribution of key variables, particularly *netinc* and *sal*, is examined using descriptive statistics and histograms to assess skewness. These variables exhibit substantial variation and potential right-skewness, which may affect regression estimates.

To address this, logarithmic transformations of netinc and sal are applied, and the regression model is re-estimated to evaluate whether the transformation improves model fit or significance.

A correlation matrix is also constructed to examine pairwise relationships among the variables.

3. Results and Recommendations:

3.1 Descriptive Statistics:

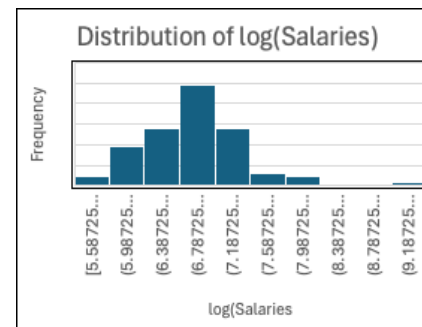
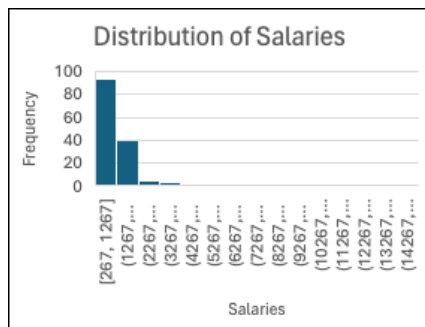
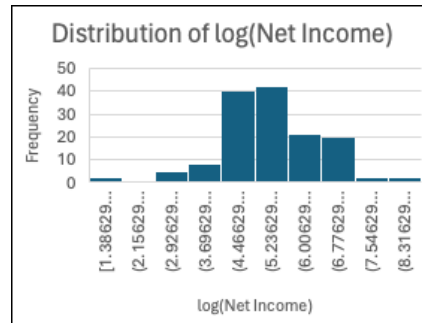
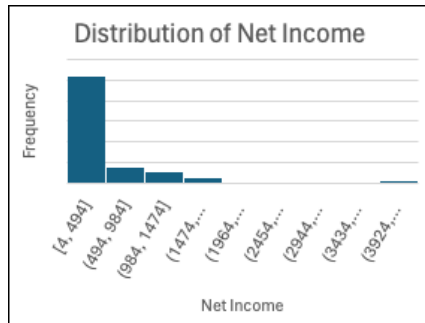
Table 1: Descriptive Statistics

Variable	Mean	Median	Min	Max	Std Dev
return	-4.04	-8.67	-84.89	131.84	39.42
roe	17.73	15.80	1.10	61.10	9.88
roc	13.71	12.05	1.40	43.30	7.88
dkr	25.65	26.40	0.00	79.50	16.67
eps	2.02	6.70	-89.30	266.60	43.15
netinc	512.32	229.50	4.00	4237.00	723.61
sal	1325.11	1040.50	267.00	14336.00	1539.89

Stock returns range from -84.89% to 131.84%, with a mean of -4.04% and a standard deviation of 39.42%, indicating wide dispersion in performance across the sample.

The profitability measures, roe and roc, are relatively more stable, while eps displays large variability because of extreme negative and positive values.

3.2 Distribution Analysis and Log Transformations:



Histograms of netinc and sal show clear right-skewed distributions, with most observations clustered at lower values and a long right tail. This may affect the normality of residuals and the efficiency of OLS estimates.

To address this skewness, logarithmic transformations of netinc and sal are applied. The transformed variables exhibit more symmetric distributions, indicating that the log transformation reduces skewness, making the variables more suitable for regression analysis. The model is re-estimated using the transformed variables to evaluate whether this improves model fit and statistical significance. Log transformation is not appropriate for dkr and eps.

The variable *eps* contains negative values due to firms reporting losses, making logarithmic transformation undefined. Similarly, *dkr* takes a value of zero for some firms, which also prevents log transformation. Both variables are therefore retained in their original form. The wide variation in *eps* reflects genuine differences in firm performance rather than data issues, and is handled within the OLS framework.

3.3 Correlation Analysis:

Table 2: Correlation Matrix

Variable	<i>roe</i>	<i>roc</i>	<i>dkr</i>	<i>eps</i>	<i>netinc</i>	<i>sal</i>	<i>return</i>
roe	1.000						
roc	0.903	1.000					
dkr	-0.254	-0.504	1.000				
eps	0.520	0.519	-0.140	1.000			
netinc	0.160	0.035	0.079	0.098	1.000		
sal	0.146	0.142	-0.023	0.124	0.183	1.000	
return	-0.143	-0.114	0.119	0.036	-0.054	0.122	1.000

Most pairwise correlations among the explanatory variables are low to moderate. However, *roe* and *roc* exhibit a very high positive correlation (0.903), indicating potential multicollinearity, as both capture similar aspects of profitability.

Apart from the *roe*–*roc* pair, no other variable combinations show strong correlations, suggesting that multicollinearity is not a widespread concern. Correlations between each explanatory variable and stock returns are uniformly weak, providing preliminary evidence that no strong linear relationship exists between the firm characteristics considered and subsequent returns.

3.4 Regression Results:

Table 3: Model Fit Statistics (Baseline Model)

Metric	Value
R ²	0.074
Adjusted R ²	0.033
F-statistic	1.798
p-value (F-test)	0.104

Table 4: OLS Regression Results (Baseline Model)

Variable	Coefficient	Std Error	p-value
Intercept	-6.707	13.071	0.609
roe	-1.660	0.958	0.086
roc	1.317	1.351	0.331
dkr	0.402	0.272	0.142
eps	0.115	0.090	0.203
netinc	-0.003	0.005	0.587
sal	0.004	0.002	0.097

The baseline regression yields an R² of 0.074 and an adjusted R² of 0.033, indicating that the model accounts for only a small proportion of the variation in stock returns. The F-test is not statistically significant at the 5% level ($p = 0.104$), implying that the explanatory variables are not jointly significant. Thus, we fail to reject the joint null hypothesis that all slope coefficients are zero. Economically, firm characteristics do not provide a reliable basis for predicting stock returns.

Table 5: Model Fit Statistics (Log-Transformed Model)

Metric	Value
R ²	0.065
Adjusted R ²	0.023
F-statistic	1.564
p-value (F-test)	0.162

Table 6: OLS Regression Results (Log-Transformed Model)

	Coefficients	Standard Error	P-value
Intercept	-41.240	40.048	0.305
roe	-1.677	0.962	0.084
roc	1.396	1.342	0.300
dkr	0.414	0.273	0.132
eps	0.127	0.091	0.166
log_netinc	-2.771	3.491	0.429
log_sal	7.543	6.277	0.232

The log-transformed model does not improve explanatory performance. The R² decreases to 0.065, and the F-test becomes less significant ($p = 0.162$), reinforcing the lack of joint significance.

Table 7: Comparison of Model Fit

Metric	Base Model	Log Model
R ²	0.074	0.065
Adjusted R ²	0.033	0.023
F-statistic	1.798	1.564
F-test p-value	0.104	0.162

Table 8: Comparison of Coefficients and p-values

Variable	Base Coeff	Base p-value	Log Coeff	Log p-value
----------	------------	--------------	-----------	-------------

Intercept	-6.707	0.609	-41.240	0.305
roe	-1.660	0.086	-1.677	0.084
roc	1.317	0.331	1.396	0.300
dkr	0.402	0.142	0.414	0.132
eps	0.115	0.203	0.127	0.166
netinc / log(netinc)	-0.003	0.587	-2.771	0.429
sal / log(sal)	0.004	0.097	7.543	0.232

As shown in Tables 7 and 8, the log transformation does not improve model fit - the baseline marginally outperforms across all metrics, and coefficient estimates remain consistent across both specifications.

At the 5% significance level, all six individual null hypotheses - $H_0: \beta_i = 0$ for each explanatory variable are corroborated. No variable achieves statistical significance in either the baseline or the log-transformed model, meaning the data provides no sufficient evidence to conclude that any of the firm characteristics examined are systematically related to stock returns.

At the 10% level, roe ($p = 0.086$) and sal ($p = 0.097$) show marginal evidence, but this falls short of the chosen significance threshold. Correspondingly, none of the individual alternative hypotheses are corroborated. The joint null hypothesis is similarly corroborated in both model specifications, with F-test p-values of 0.104 and 0.162 respectively, confirming that the explanatory variables collectively possess no predictive power over stock returns.

Economically, the insignificance of roe and roc among the most widely tracked measures of firm performance implies that investors could not have earned superior returns by selecting high-profitability firms based on 2014 data, consistent with this information already being reflected in prices. The insignificance of eps and netinc similarly indicates that earnings-based information offered no reliable predictive advantage. The near-significance of sal ($p = 0.097$) weakens further in the log-transformed model and falls short of the 5% threshold, making it insufficient to draw a firm conclusion. Overall, no variable in this dataset constitutes a meaningful

signal for stock selection, reinforcing that these publicly available characteristics were efficiently priced by the market.

3.5 Recommendations:

Investors should avoid relying solely on firm-level indicators such as profitability measures, earnings, leverage, and CEO compensation when making stock selection decisions. Instead, a more comprehensive approach that incorporates broader information, such as market-wide conditions (e.g., interest rates, economic growth), valuation metrics (e.g., price-to-earnings and price-to-book ratios), and industry-specific factors may provide a more reliable basis for evaluation.

4. Conclusion and Limitations:

The results of this study provide no evidence in support of return predictability. Both models exhibit low explanatory power and lack joint significance. Thus, firm-level characteristics do not explain stock returns over the period. These findings are broadly consistent with the Efficient Market Hypothesis, which states that publicly available information is already reflected in stock prices.

However, the analysis is subject to several limitations. The sample size is restricted to 142 firms and a single time period, limiting the generalizability of the results. The model includes only a limited set of firm-specific variables, and omitted variable bias may be present. Additionally, the high correlation between roe and roc indicates potential multicollinearity, which may affect estimates.

Future research could address these limitations by expanding the dataset, incorporating additional variables, and exploring alternative econometric approaches.

5. References:

Fama, Eugene F. “Efficient Capital Markets: A Review of Theory and Empirical Work.” *The Journal of Finance*, vol. 25, no. 2, 1970, pp. 383–417.

Wooldridge, Jeffrey M. *Introductory Econometrics: A Modern Approach*. 7th ed., Cengage Learning, 2020.

EM>ReturnsData.xlsx. Course-Provided Dataset. ECO348-FAC413-DOM637.

LP2 Project Brief. Course Instructions. ECO348-FAC413-DOM637.